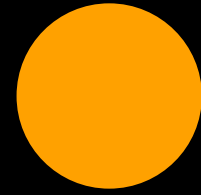


Final Presentations

SUPPLY CHAIN ANALYTICS

By Varsha Gunturu



DATA SET



[Browse](#)

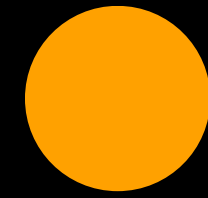
Q

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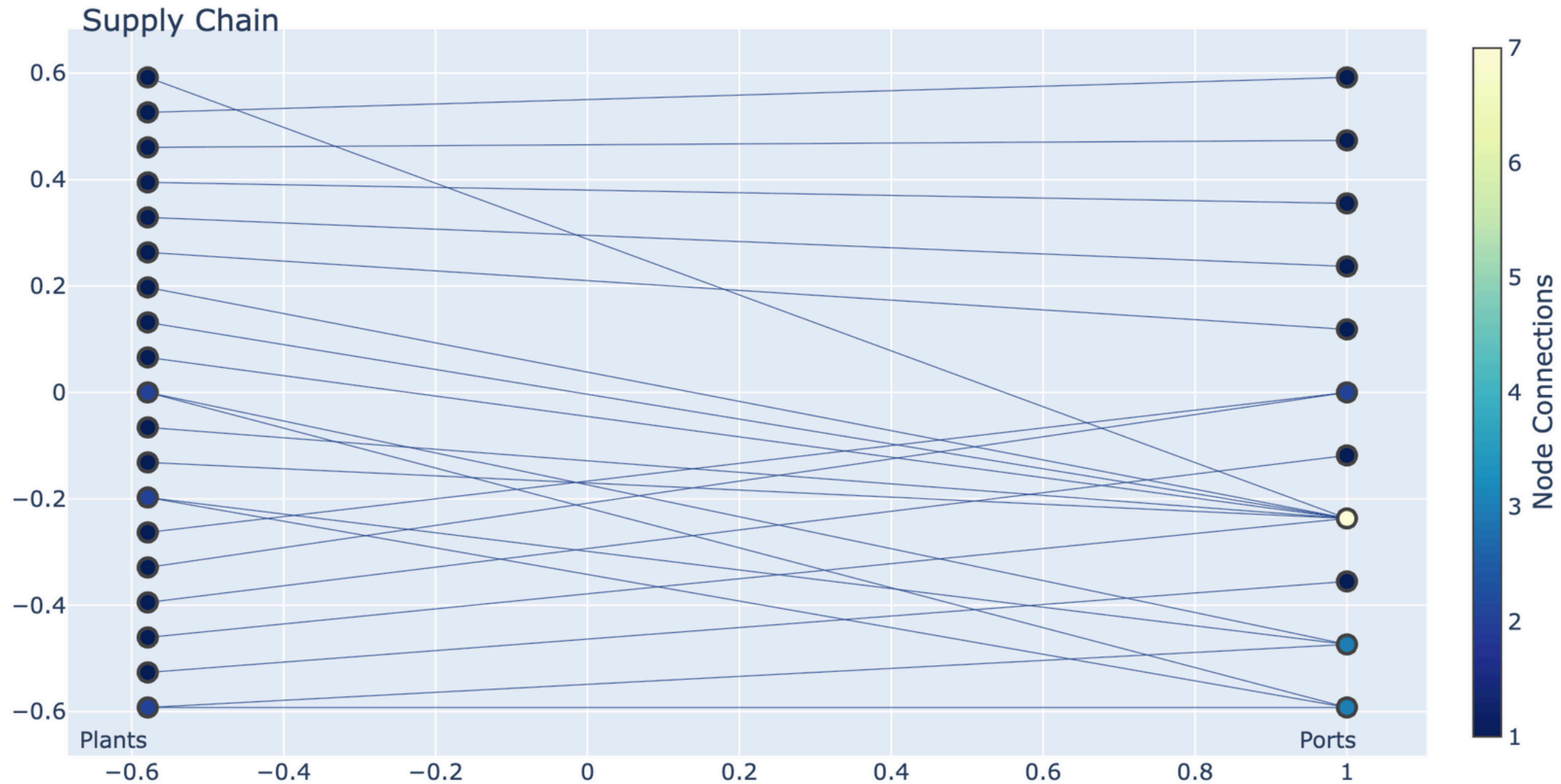
	A	B	C	D	E	F	G	H	I	J	
1	Order ID	Order Date	Origin Port	Carrier	TPT	Service Level	Ship ahead day count	Ship Late Day count	Customer	Product ID	
2	1447296446.7	41420	PORT09	V44_3	1	CRF	3	0	V55555_53	1700106	
3	1447158014.7	41420	PORT09	V44_3	1	CRF	3	0	V55555_53	1700106	
4	1447138898.7	41420	PORT09	V44_3	1	CRF	3	0	V55555_53	1700106	
5	1447363527.7	41420	PORT09	V44_3	1	CRF	3	0	V55555_53	1700106	
6	1447363980.7	41420	PORT09	V44_3	1	CRF	3	0	V55555_53	1700106	
7	1447351440.7	41420	PORT09	V44_3	1	CRF	3	0	V55555_53	1700106	
8	1447320235.7	41420	PORT09	V44_3	1	CRF	3	0	V55555_53	1700106	
9	1447158018.7	41420	PORT09	V44_3	1	CRF	3	0	V55555_53	1700106	
10	1447219340.7	41420	PORT09	V44_3	1	CRF	3	0	V55555_53	1700106	
11	1447398415.7	41420	PORT09	V44_3	1	CRF	3	0	V55555_53	1700106	
OrderList		FreightRates	WhCosts	WhCapacities	ProductsPerPlant	VmiCustomers	PlantPorts				

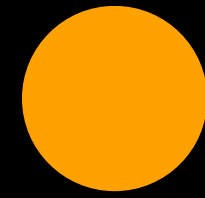
The dataset was taken from Brunel University London with 7 sheets detailing one day's order of a company.

The order list contained 9216 records and the other sheets contained information on plants, products, ports, warehouse and other costs.

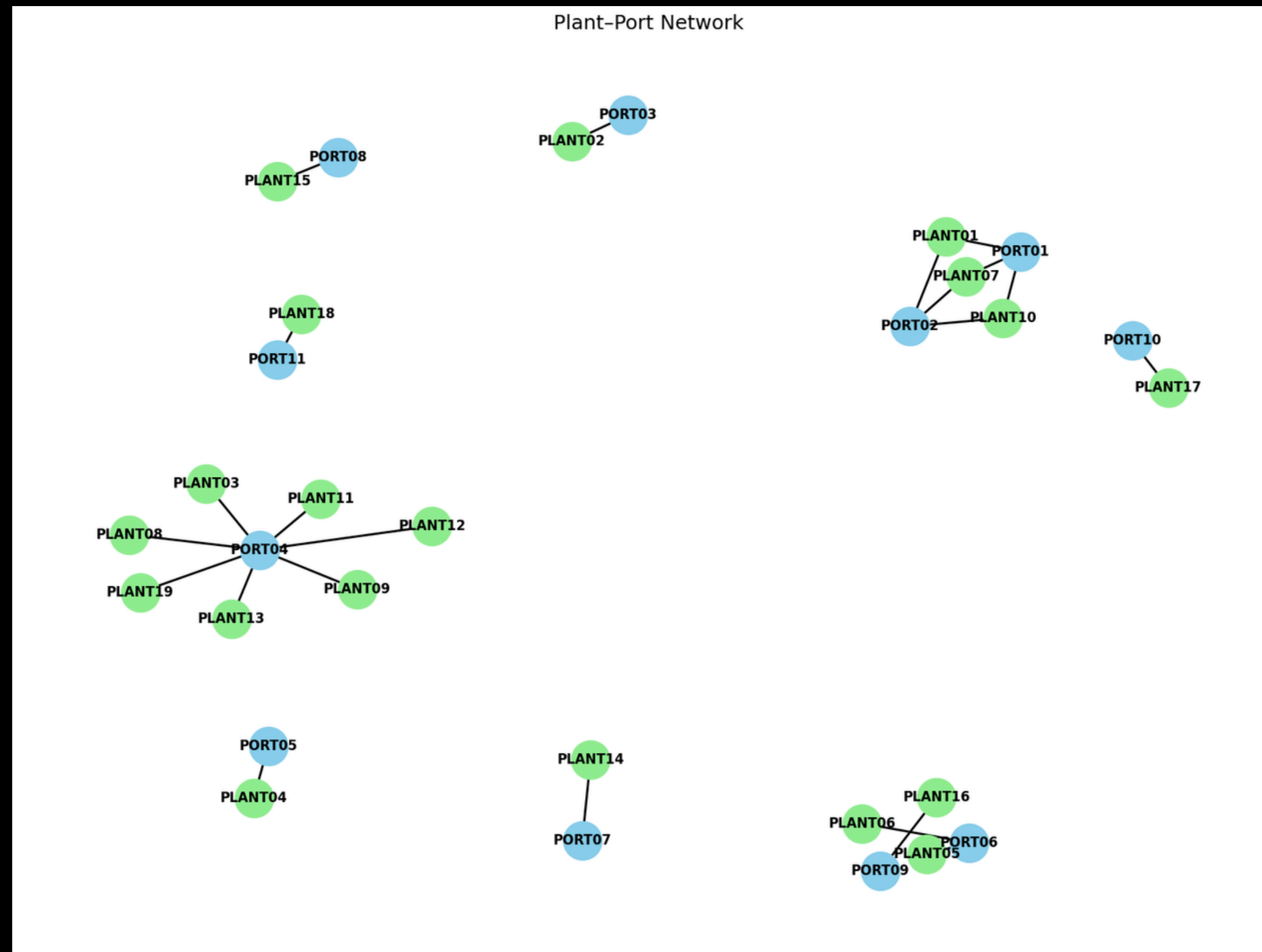


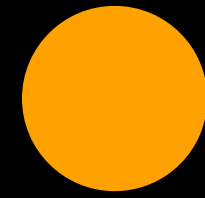
EDA ON DATASET



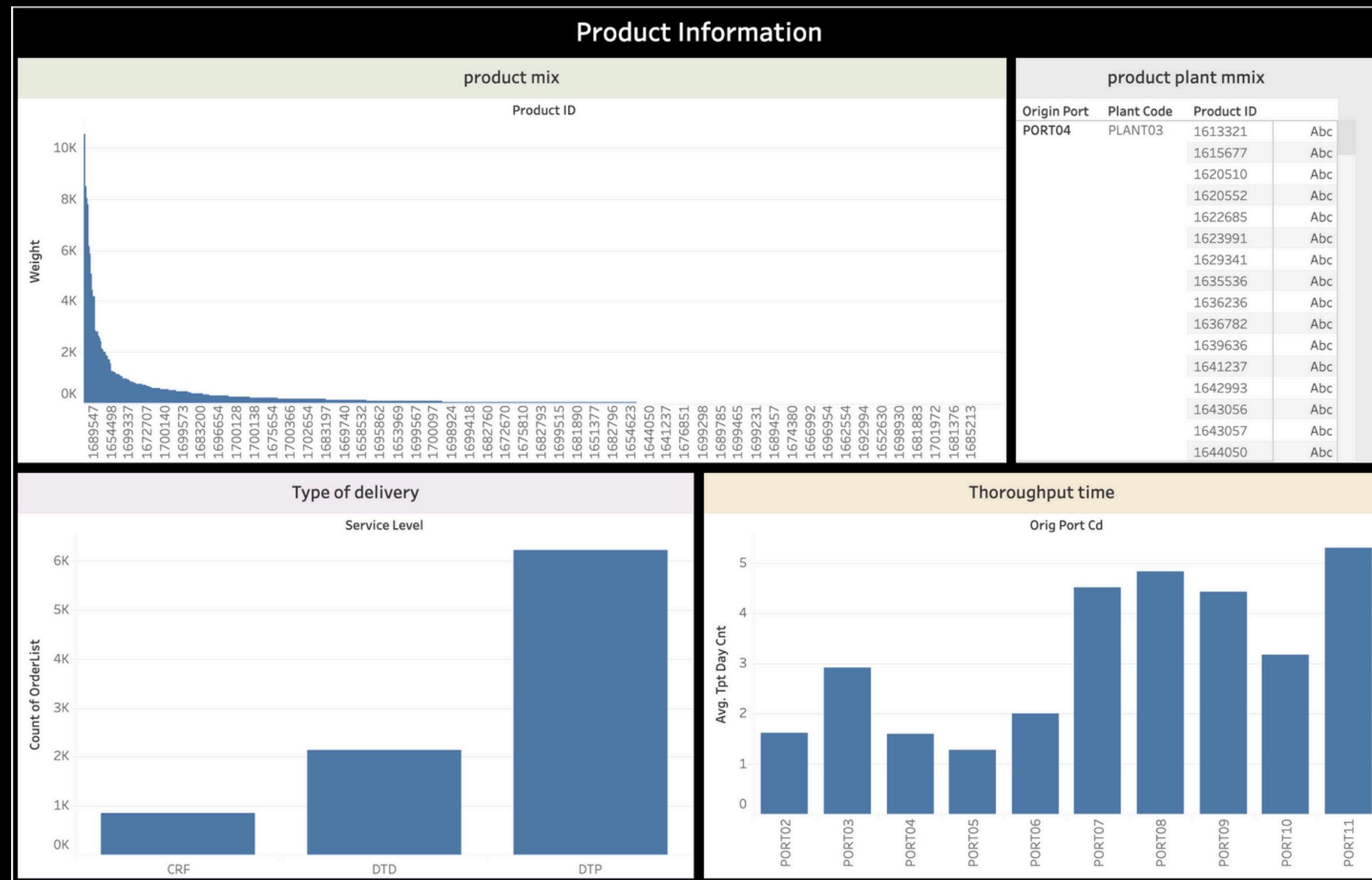


EDA ON DATASET

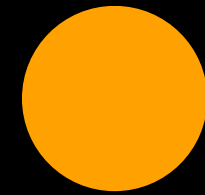




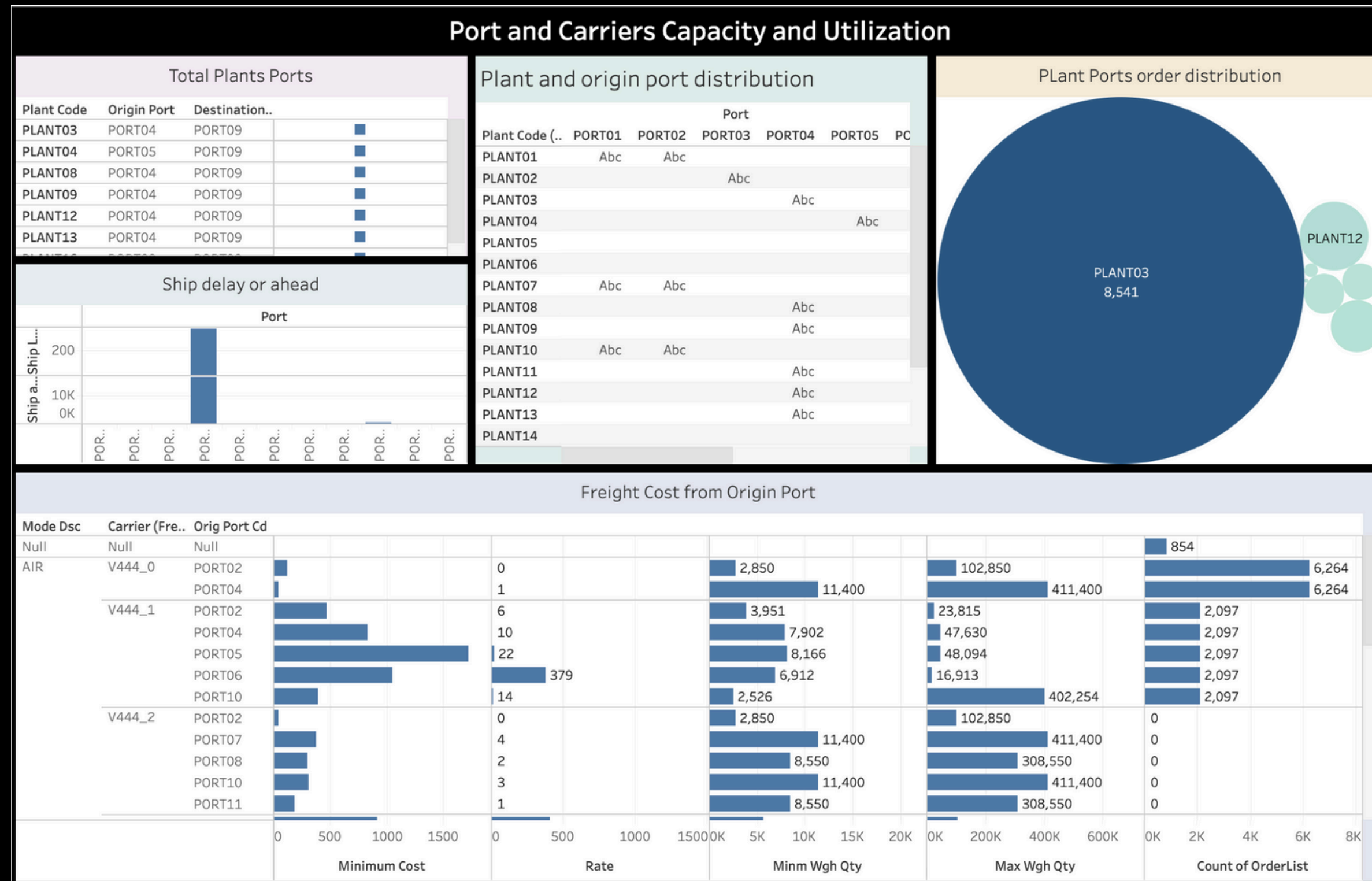
EDA ON DATASET



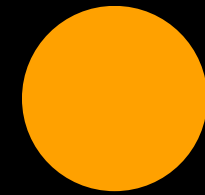
Link: [Tableau public](#)



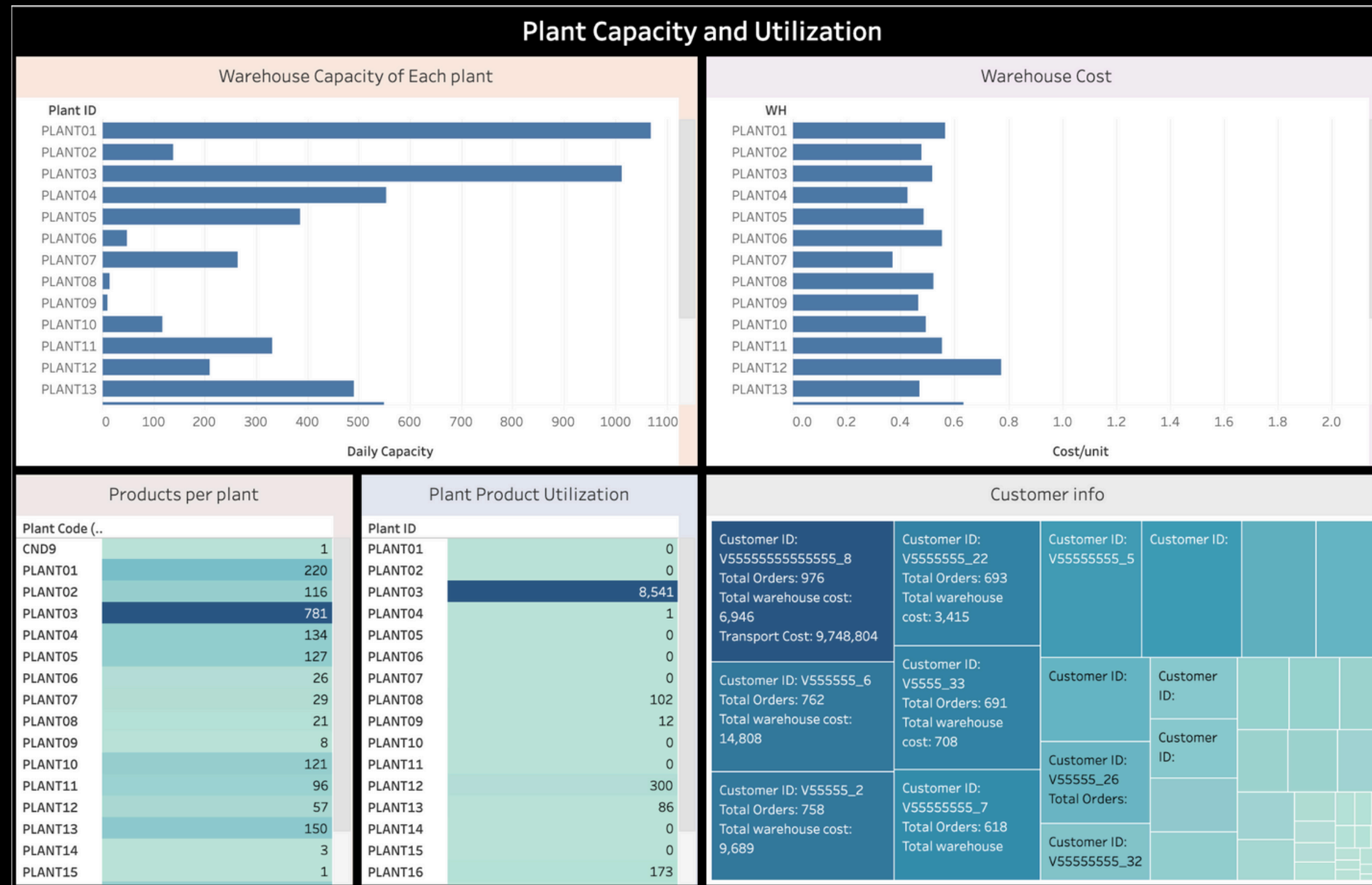
EDA ON DATASET



Link: [Tableau public](#)



EDA ON DATASET



Link: [Tableau public](#)

PROBLEM

Problem:

- To reduce the burden on a single plant and at the same time optimize the production and cost
- The demand of each individual product is way higher than total capacity of warehouse
- warehouse capacity: 5792, order demand: 9216, (using dummy plants to manage extra)

Constraints:

- Each plant has a finite warehouse capacity, by capacity meaning orders that can be handled irrespective of product type.
 - Assumption: X customer has orders 12 units of product A and 13 units of product B, the total order of customer X is considered as 1 warehouse capacity
- Each plant doesn't produce all products: Certain plants produce certain products
- Certain plants only produce for certain customers called vmicustomers
- Each plant is connected to one or two ports
- All destination port is the same which is the PORT9.
- Freight cost with minimum and maximum weight limits
 - Some CRF customers handle their own transport so their freight cost is 0.

SOLUTION

Multiple Integer Linear programming using python

Strategy: Clubbing orders from same customers into one as long as they have the same combination of products all produced by one plant and within the freight weight min and max limitation. Selecting the one overall least cost.

```
Welcome to the CBC MILP Solver
Version: 2.10.3
Build Date: Dec 15 2019

command line - /Users/gunturuvarsha/Library/Python/3.9/lib/python/site-packages/pulp/apis/./solverdir/cbc/osx/i64/cbc /V
At line 2 NAME          MODEL
At line 3 ROWS
At line 29740 COLUMNS
At line 154801 RHS
At line 184537 BOUNDS
At line 205734 ENDDATA
Problem MODEL has 29735 rows, 21196 columns and 62162 elements
Coin0008I MODEL read with 0 errors
Option for timeMode changed from cpu to elapsed
Continuous objective value is 1.55167e+07 - 0.04 seconds
Cgl0004I processed model has 0 rows, 0 columns (0 integer (0 of which binary)) and 0 elements
Cbc3007W No integer variables - nothing to do
Cuts at root node changed objective from 1.55167e+07 to -1.79769e+308
Probing was tried 0 times and created 0 cuts of which 0 were active after adding rounds of cuts (0.000 seconds)
Gomory was tried 0 times and created 0 cuts of which 0 were active after adding rounds of cuts (0.000 seconds)
Knapsack was tried 0 times and created 0 cuts of which 0 were active after adding rounds of cuts (0.000 seconds)
Clique was tried 0 times and created 0 cuts of which 0 were active after adding rounds of cuts (0.000 seconds)
MixedIntegerRounding2 was tried 0 times and created 0 cuts of which 0 were active after adding rounds of cuts (0.000 seconds)
FlowCover was tried 0 times and created 0 cuts of which 0 were active after adding rounds of cuts (0.000 seconds)
TwoMirCuts was tried 0 times and created 0 cuts of which 0 were active after adding rounds of cuts (0.000 seconds)
...
Total time (CPU seconds):      0.22   (Wallclock seconds):      0.28

Status: Optimal
Optimal total cost: 15516670.860201446
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

VALIDATION

NO of orders assigned to dummy
plant: 0

	Used_Capacity	Max_Capacity	Utilization_%
Plant			
PLANT09	10	11.0	90.909091
PLANT08	5	14.0	35.714286
PLANT12	13	209.0	6.220096
PLANT03	41	1013.0	4.047384
PLANT02	5	138.0	3.623188
PLANT04	20	554.0	3.610108
PLANT05	12	385.0	3.116883
PLANT18	2	111.0	1.801802
PLANT13	8	490.0	1.632653
PLANT07	4	265.0	1.509434
PLANT16	4	457.0	0.875274
PLANT10	1	118.0	0.847458
PLANT01	2	1070.0	0.186916
PLANT14	1	549.0	0.182149
DUMMY	46	NaN	NaN

	OrderIndex	Order ID	Customer	Product ID	Service Level	Destination Port	Assigned Plant	Units	Weight	Total Cost (this order, this plant)
0	1193	1.447327e+09	V5555555555555555_29	1688629	DTP	PORT09	PLANT02	1807	134.523983	882.213259
1	1250	1.447327e+09	V5555555555555555_29	1689547	DTP	PORT09	PLANT02	1253	77.986272	617.676309
2	2951	1.447159e+09	V5555555555_3	1688587	DTP	PORT09	PLANT02	529	17.515149	271.963762
3	2955	1.447326e+09	V5555555555_3	1688587	DTP	PORT09	PLANT02	494	19.241526	255.251138
4	2956	1.447259e+09	V5555555555_3	1688575	DTP	PORT09	PLANT02	474	18.086819	245.701068

Baseline total cost: 15648713.914530035

Optimized total cost: 15516670.860201446

Savings: 132043.05432858877

Savings %: 0.8437949281313467

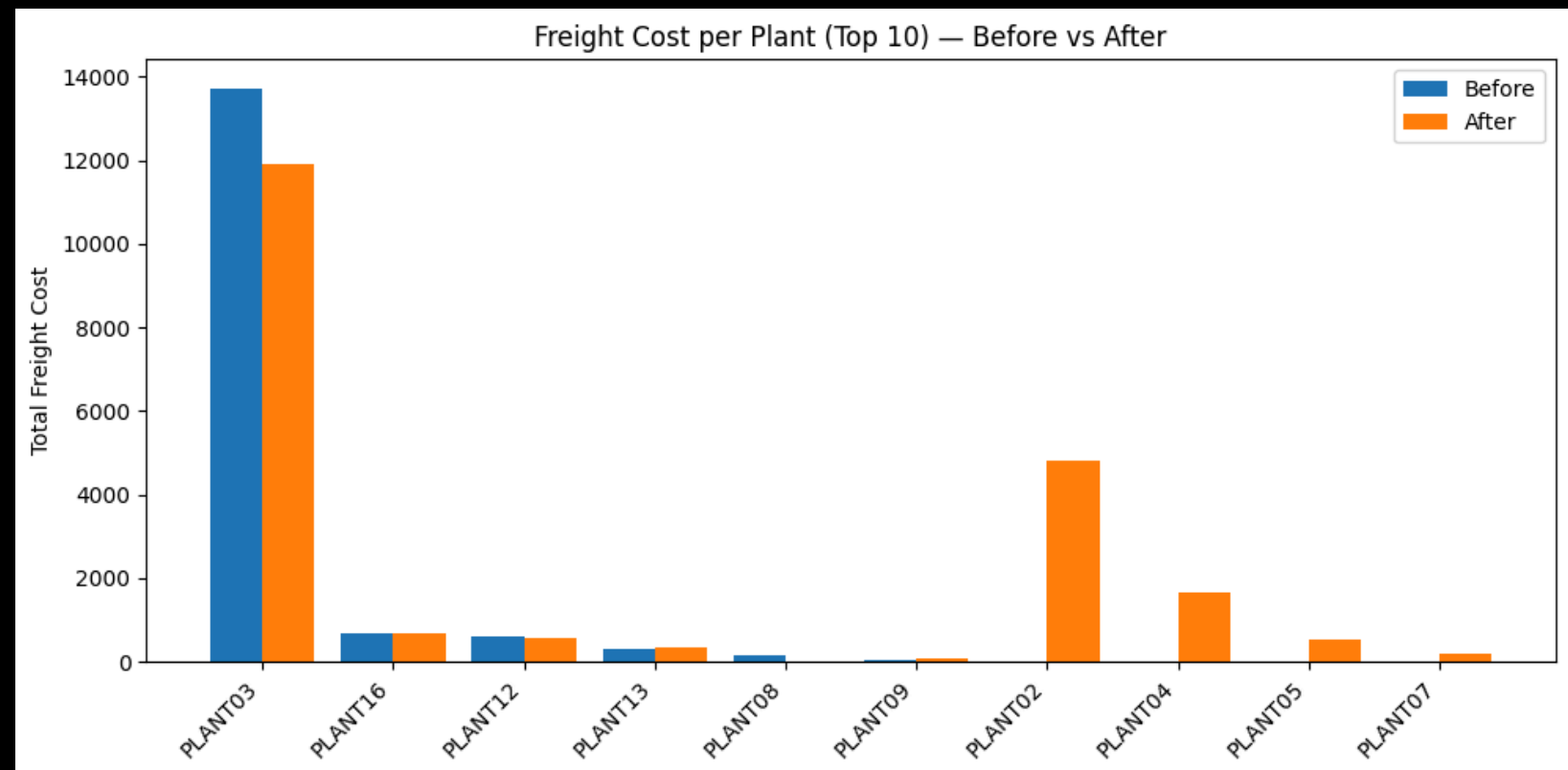
ORDERS PER PLANT – BEFORE OPTIMIZATION			
Plant Code	Num_Orders	Total_Units	Total_Weight
PLANT03	8541	28611617	135486.511563
PLANT12	300	112000	8147.429882
PLANT16	173	245133	32168.011913
PLANT08	102	277008	1478.787971
PLANT13	86	43379	5428.393538
PLANT09	12	223830	406.364549
PLANT04	1	348	2.101107

ORDERS PER PLANT – AFTER OPTIMIZATION				
Assigned Plant	Num_Orders	Total_Units	Total_Weight	Total_Cost
PLANT03	7482	26973428	115829.007628	1.397072e+07
PLANT04	387	487665	7107.532341	2.106333e+05
PLANT12	288	105196	7160.348996	8.190210e+04
PLANT02	249	188235	9474.763017	9.470461e+04
PLANT10	181	66834	1378.902448	3.331451e+04
PLANT05	180	248179	1015.543965	1.216592e+05
PLANT16	173	245133	32168.011913	4.712767e+05
PLANT07	102	276732	1479.231543	1.029705e+05
PLANT13	98	50183	6415.474425	2.391484e+04
PLANT09	74	870644	1086.245337	4.050049e+05
PLANT08	1	1086	2.538912	5.693222e+02

VALIDATION

	Before_Num_Orders	Before_Units	Before_Weight	After_Num_Orders	After_Units	After_Weight	After_Cost	Change_in_Orders	Change_in_Units
PLANT04	1	348.0	2.101107	387	487665	7107.532341	2.106333e+05	386	487317.0
PLANT02	0	0.0	0.000000	249	188235	9474.763017	9.470461e+04	249	188235.0
PLANT10	0	0.0	0.000000	181	66834	1378.902448	3.331451e+04	181	66834.0
PLANT05	0	0.0	0.000000	180	248179	1015.543965	1.216592e+05	180	248179.0
PLANT07	0	0.0	0.000000	102	276732	1479.231543	1.029705e+05	102	276732.0
PLANT09	12	223830.0	406.364549	74	870644	1086.245337	4.050049e+05	62	646814.0
PLANT13	86	43379.0	5428.393538	98	50183	6415.474425	2.391484e+04	12	6804.0
PLANT16	173	245133.0	32168.011913	173	245133	32168.011913	4.712767e+05	0	0.0
PLANT12	300	112000.0	8147.429882	288	105196	7160.348996	8.190210e+04	-12	-6804.0
PLANT08	102	277008.0	1478.787971	1	1086	2.538912	5.693222e+02	-101	-275922.0
PLANT03	8541	28611617.0	135486.511563	7482	26973428	115829.007628	1.397072e+07	-1059	-1638189.0

	Freight_Before	Freight_After	Delta_Freight
PLANT02	0.000000	4821.735600	4821.735600
PLANT03	13727.818717	11920.786306	-1807.032411
PLANT04	0.000000	1667.288144	1667.288144
PLANT05	0.000000	511.970713	511.970713
PLANT07	0.000000	185.614642	185.614642
PLANT08	153.426099	1.499200	-151.926899
PLANT09	26.940086	93.581640	66.641554
PLANT10	0.000000	326.455481	326.455481
PLANT12	620.795469	571.727041	-49.068428
PLANT13	294.462663	343.531091	49.068428
PLANT16	668.556730	668.556730	0.000000



VALIDATION

Total warehouse cost BEFORE: 15633221.914766343

Total warehouse cost AFTER : 15495558.113614047

WH Savings : 137,663.80115229636

WH Savings % : 0.8805849613269173

Total freight before: 15491.999763692163

Total freight after : 21112.74658747506

Savings : -5620.746823782896

Savings % : -36.28160927910652

Transport is optimized but is not used well hence why transport cost increased, whereas overall cost decreases with approximately 0.84% of savings.

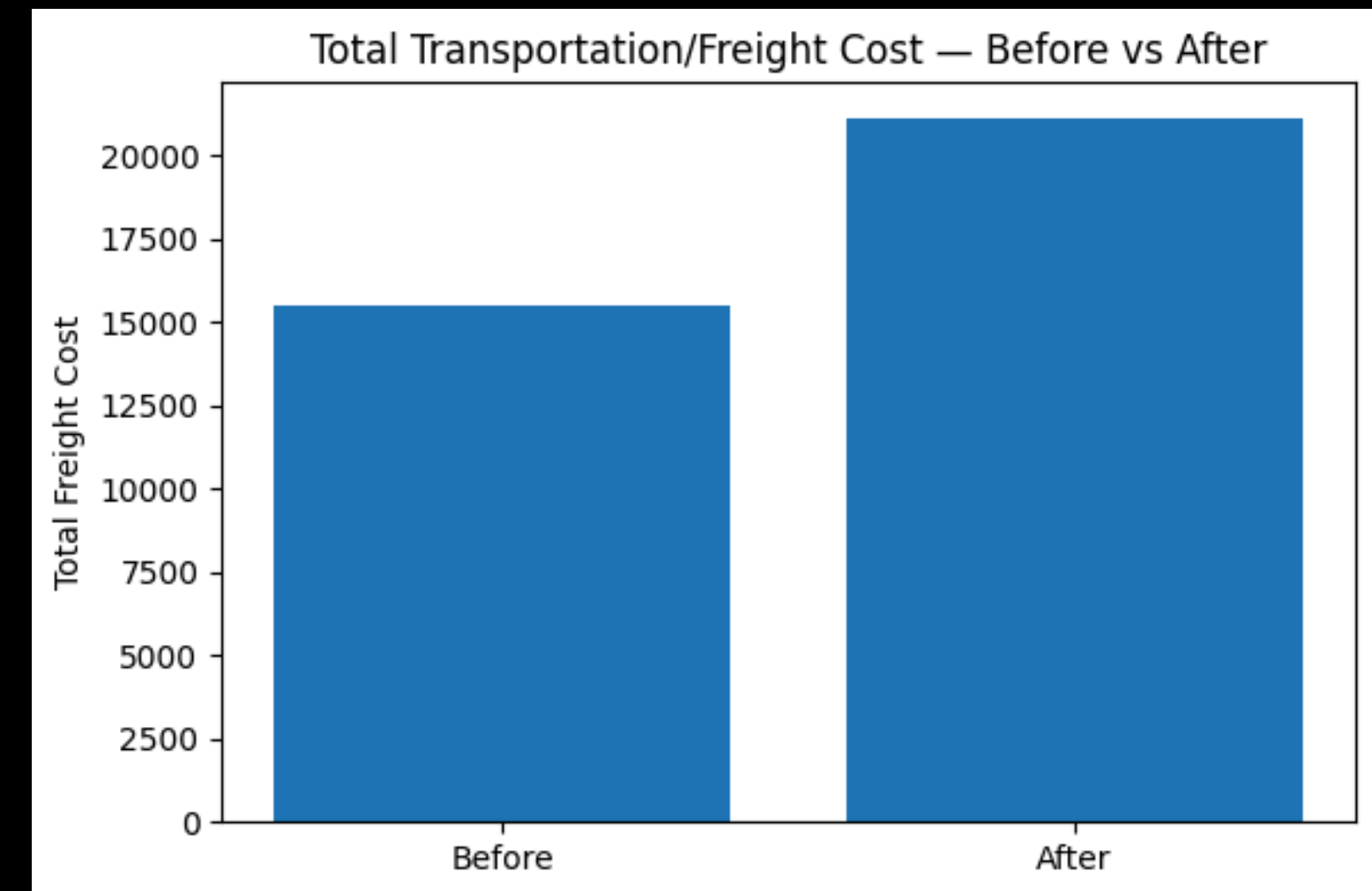
=== Transport-only optimization (plants fixed) ===

Total baseline freight cost : 15491.999763692165

Total optimized freight cost: 15491.999763692165

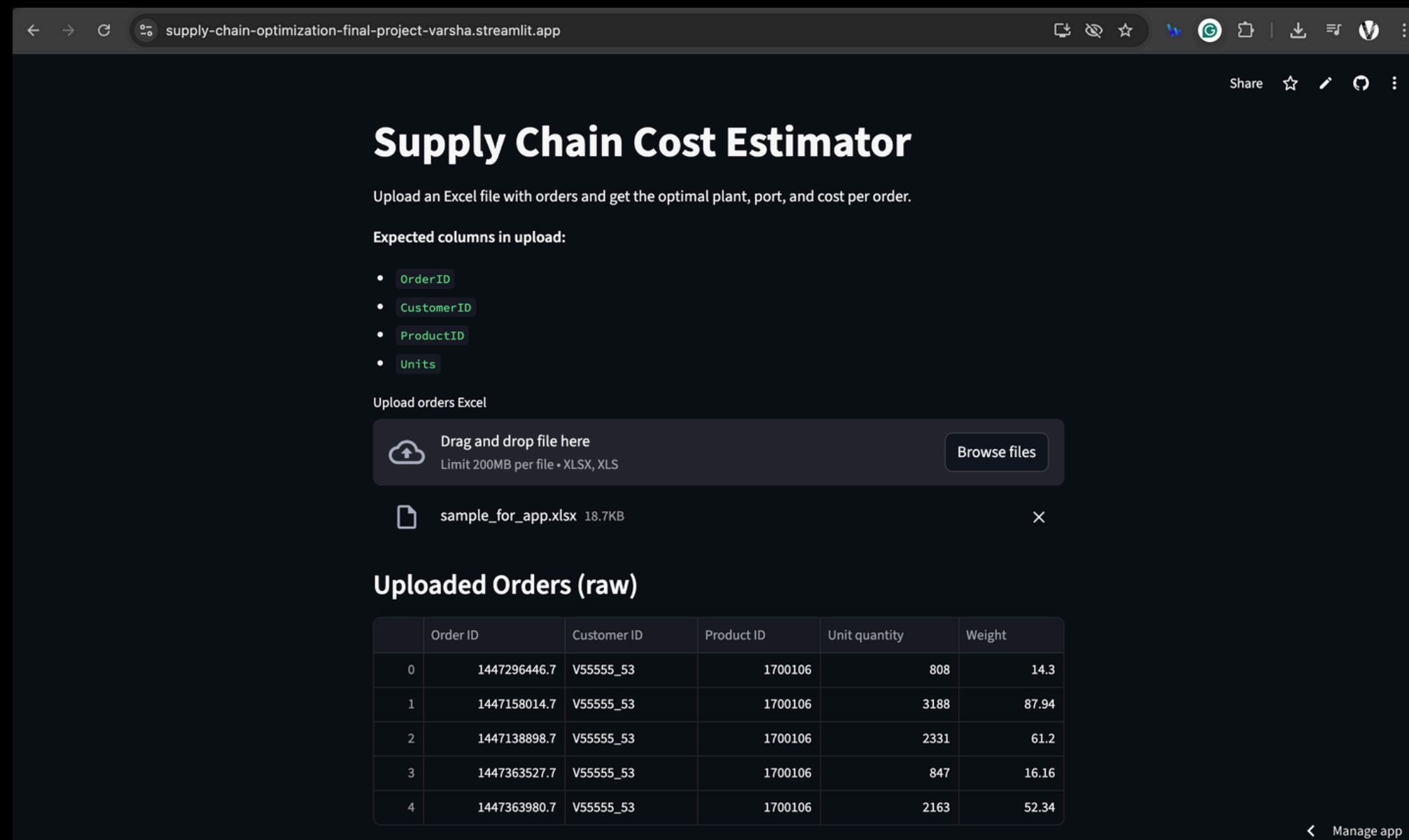
Freight savings : 0.0

Freight savings % : 0.0



APP

Deployed Streamlit app



Cost Summary by Plant

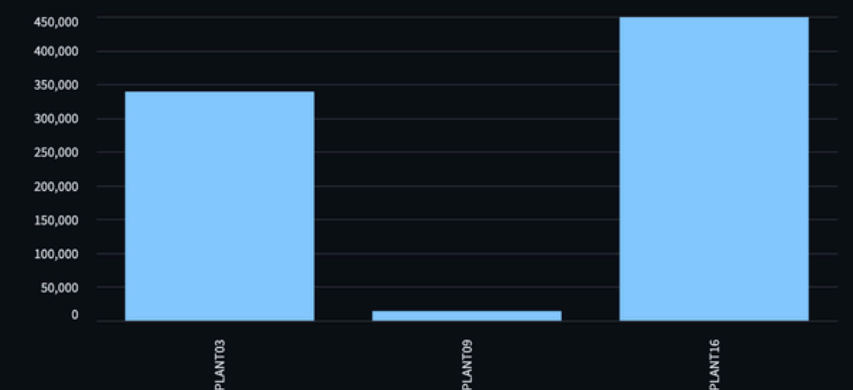
AssignedPlant	Total_Units	Total_Weight	Total_Freight	Total_WH	Total_Cost	Num_Orders
PLANT16	232681	27614.73	2502.6379	446702.7307	449205.3686	139
PLANT03	654633	1530.2741	192.8845	338773.8158	338966.7004	123
PLANT09	30472	52.1639	8.9952	14171.6465	14180.6417	6

Supply Chain Network Visualization

Optimized Supply Chain Route (Minimal Used Network)



Total Cost by Plant (Bar Chart)



Total Freight vs WH Cost (Overall)

Total Freight Cost

2,705

Total Warehouse Cost

799,648



THANK YOU